

Sameer Shukla

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PROFILE

Research-focused Machine Learning Engineer with expertise in Computer Vision, Document Intelligence, and Deep Learning. Proven ability to optimize architectures like LayoutLMv3 and EfficientNet for high-accuracy inference. Passionate about developing novel algorithms and scalable pipelines for multimodal learning systems.

EDUCATION

Bennett University

B.Tech - Computer Science

Greater Noida, UP

Expected June 2027

Cumulative GPA: 8.83

Relevant Coursework: Artificial Intelligence, Data Structures & Algorithms, Probability & Statistics, Software Engineering.

UNIVERSITY PROJECTS

Intelligent Document Processing (Invoice Extraction) | *PyTorch, Transformers, LayoutLMv3*

- Designed a multimodal information extraction pipeline using **LayoutLMv3** to process unstructured invoice images, achieving high F1 scores on entity recognition tasks.
- Optimized GPU inference performance on NVIDIA Graphic Card by experimenting with **mixed-precision training (FP16)**, reducing processing time by **35%** without accuracy loss.
- Conducted ablation studies on transformer attention heads to handle noisy real-world document layouts, improving field extraction accuracy for complex tables.

Krishi Kawach (Agricultural AI Research) | *TensorFlow, Keras, EfficientNet, Flask*

Github

- Architected a custom transfer learning framework based on **EfficientNet-B3**, achieving **97% validation accuracy** in disease identification by implementing aggressive data augmentation and class-balancing techniques.
- Engineered an end-to-end inference pipeline integrating Flask with model serialization, reducing latency for real-time disease diagnosis on mobile edge cases.
- Formulated a comparative analysis of CNN architectures, selecting the optimal backbone based on a trade-off between floating-point operations (FLOPs) and recall rates.

Pashu Mitra (Livestock Biometrics) | *YOLOv8, OpenCV, EfficientNet*

Github

- Developed a cascading object detection system using YOLO for localization and EfficientNet-B2/B3 for fine-grained species classification.
- Optimized the region-proposal strategy, applying classifiers strictly to cropped bounding boxes to lower computational overhead by **~40%**.
- Implemented a robust data ingestion pipeline to handle diverse livestock imagery, significantly reducing false positive rates in real-world testing scenarios.

TECHNICAL SKILLS

- Core Research Areas:** Computer Vision, Document Intelligence, Deep Learning, Pattern Recognition.
- Languages:** Python (Advanced), SQL, C++, JavaScript.
- Frameworks & Libraries:** PyTorch, Hugging Face Transformers, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas.
- Tools & Platforms:** Git, Docker, Flask, Vercel, Terraform, Google Colab, VS Code.

CERTIFICATIONS & ACHIEVEMENTS

- Certifications:** Machine Learning Specialization (DeepLearning.AI), Software Engineering (IIT Kharagpur/NPTEL), Data Visualization with R (IBM).
- Awards:** Smart India Hackathon (SIH) - Cleared internal round with AI-based agricultural solution (Aug 2024).